Supplementary material: From local quantum-field measurements to certified resources

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Equations numbered S1, S2, and so on belong to this supplement; unprefixed cross-references point to the main article. References are numbered separately in each document.

S1 Matched benchmark for comparing measurement frameworks

The approaches discussed above should be compared on a common operational task, rather than by placing their formal expressions side by side. We specify a benchmark that can be implemented first for a free real scalar field in four-dimensional Minkowski spacetime and then extended to curved backgrounds or gauge fields. The benchmark is deliberately modest: it does not assert that QTP, detector instruments, the positive formalism, and local S-matrix constructions are equivalent. It identifies which predictions can be matched and which additional structures each framework supplies.

Let $\hat{\phi}(x)$ be a free scalar field in the Minkowski vacuum $|0\rangle$, and let two initially uncorrelated two-level probes A and B have gaps Ω_A and Ω_B. The initial joint state is $|0\rangle\langle 0|\otimes\rho_A\otimes\rho_B$. Their interaction regions are compact spacetime sets $\mathcal{O}_A$ and $\mathcal{O}_B$. For a finite-size probe, introduce real smooth test functions $F_\nu(x)$ supported in $\mathcal{O}_\nu$, with $\nu = A, B$. A convenient interaction-picture coupling is

$$H_I(t) = \lambda \sum_{\nu=A,B} \chi_\nu(t)\, \hat{\mu}_\nu(t)\, \hat{\Phi}_\nu(t), \qquad \hat{\Phi}_\nu(t) = \int d^3\mathbf{x}\, f_\nu(t,\mathbf{x})\hat{\phi}(t,\mathbf{x}), \tag{S1}$$

where $\hat{\mu}_\nu(t) = \hat{\sigma}_\nu^+ e^{i\Omega_\nu t} + \hat{\sigma}_\nu^- e^{-i\Omega_\nu t}$, and the spacetime support of $\chi_\nu f_\nu$ is contained in $\mathcal{O}_\nu$. Equivalently, the coupling can be written using a single spacetime test function F_ν. The field input, probe preparation, smearing, switching, coupling strength, and readout are held fixed when changing the measurement framework.

For one probe initially in its ground state, the leading excitation probability is

$$P_\nu^{(2)} = \lambda^2 \int d^4x\, d^4x'\, F_\nu(x) F_\nu(x')\, e^{-i\Omega_\nu[t(x)-t(x')]} W_0(x,x'), \tag{S2}$$

up to the convention used to absorb the switching and spatial profile into F_ν. Here $W_0(x,x') = \langle 0|\hat{\phi}(x)\hat{\phi}(x')|0\rangle$ is the Wightman function. Equation (S2) is the common reference quantity for the first comparison; a detector model may add a finite readout window, internal noise, or a nontrivial response function.

In QTP, the corresponding primary output is a time-resolved event probability density. The detector response kernel weights the same unequal-time field correlator, and an appropriate integration over the recorded detection time and detector response should recover the total transition probability in the regime where the QTP model reduces to the chosen detector coupling [1, 2]. This is a matching statement about a specified response model, not an identity between all QTP prescriptions and all Unruh–DeWitt detectors. The advantage of QTP is

that it retains the detection-time variable and generalizes naturally to joint event probabilities involving higher-point unequal-time correlators.

The positive-formalism construction addresses a different part of the benchmark. It supplies a probabilistically consistent and compositional description of localized measurements, including a specified class of quadratic observables, and makes locality and composition explicit [3, 4]. In the Oeckl–Zampeli construction, however, a modulus-square probe for several observables encodes a joint expectation rather than a joint distribution of simultaneous outcomes; probabilities are obtained separately from the positive formalism's ratio rule. To compare its output with Eq. (S2), one must specify a probe and outcome functional corresponding to the same smeared observable and readout. Agreement of a single-measurement probability is a necessary check, but the distinctive test is whether two localized measurement probes compose consistently when their spacetime regions are combined or separated. The positive formalism does not, by itself, select the same two-level detector state or define its negativity.

The local S-matrix approach supplies yet another test. The same smeared coupling is used to construct local operations and, where an explicit pointer is included, field-induced Kraus operators. The question is whether the operations satisfy the relevant causal factorization conditions and whether the resulting conditional updates avoid signalling in configurations involving multiple regions [5]. This is not a second formula for the detector excitation probability: it is a test of the causal admissibility and compositional behavior of the operation that produces the record. In a time-extended pointer model, the retarded self-propagator also constrains the attainable measurement resolution, so the pointer width and readout noise must be included in a fair comparison.

The second stage uses both probes. We require the full supports $\mathcal{O}_A$ and $\mathcal{O}_B$ to be strictly spacelike separated, not merely their centers. For a free scalar field, microcausality gives

$$[\hat{\Phi}_A, \hat{\Phi}_B] = i\Delta(F_A, F_B) = 0, \qquad W(F_A, F_B) = \langle 0|\hat{\Phi}_A\hat{\Phi}_B|0\rangle \text{ need not vanish}, \tag{S3}$$

where Δ is the Pauli–Jordan commutator distribution and $W(F_A, F_B)$ denotes the correspondingly smeared Wightman function. Thus the causal response between the regions vanishes while their vacuum correlations can remain nonzero. For initially ground-state two-level probes, the relevant block of the partial transpose, in the basis $\{|eg\rangle, |ge\rangle\}$, is

$$(\rho_{AB}^{T_B})_{\{|eg\rangle,|ge\rangle\}} = \begin{pmatrix} P_A & \mathcal{M} \\ \mathcal{M}^* & P_B \end{pmatrix} + O(\lambda^4). \tag{S4}$$

Its potentially negative eigenvalue at second order is

$$\lambda_-^{(2)} = \frac{P_A + P_B - \sqrt{(P_A - P_B)^2 + 4|\mathcal{M}|^2}}{2}. \tag{S5}$$

The corresponding leading negativity is

$$\mathcal{N}^{(2)} = \max\left\{0, \frac{\sqrt{(P_A - P_B)^2 + 4|\mathcal{M}|^2} - (P_A + P_B)}{2}\right\}, \tag{S6}$$

up to higher-order corrections. Its positivity condition is $|\mathcal{M}| > \sqrt{P_A P_B}$; for identical probes, $P_A = P_B = P$, it reduces to $\mathcal{N}^{(2)} = \max\{0, |\mathcal{M}| - P\}$. The exact expression must be re-derived if the probe dimension, initial state, coupling, or perturbative order is changed.

This two-region comparison has three distinct outputs. First, QTP supplies joint detection-time probabilities, whose leading terms involve four-point unequal-time field correlators; for the free Gaussian vacuum these can be reduced by Wick's theorem, but they are still event statistics rather than a detector density matrix. Second, the specified two-probe detector–field realization supplies the final probe state and hence the negativity in Eq. (S6). Third,

the positive-formalism and local-S-matrix approaches test whether the corresponding localized operations and their composition are probabilistically consistent and causally admissible. A QTP coincidence covariance,

$$C_{AB} = p_{AB} - p_A p_B, \tag{S7}$$

must therefore not be identified with $\mathcal{N}$: a nonzero covariance is a correlation witness, not by itself an entanglement measure.

A useful numerical and conceptual comparison should report the following quantities separately: (i) the integrated single-probe excitation probability and its time-resolved distribution; (ii) the joint event probability and coincidence covariance; (iii) the final two-probe density matrix and its resource witness, when an explicit probe instrument is specified; (iv) a no-signalling residual obtained by varying the remote setting while holding the local setting fixed; and (v) the detector's dark-count probability, efficiency, and readout resolution. The same field state, compact supports, smearing functions, probe preparation, and coupling must be used in all calculations. A timelike or causally connected configuration should be included as a control, not mixed with the strictly spacelike benchmark.

The comparison can then be summarized as follows:

Framework	Primary output in the benchmark	Question it can answer directly
UDW / explicit probe instrument	Probe transition probabilities and final probe state	How much resource is transferred to the specified probe?
QTP	Time-resolved and joint event probabilities	When do events occur, and what multi-time correlations are recorded?
Positive formalism	Probabilities for localized probes and their spacetime composition	Are localized measurements positive, local, and compositional?
Local S matrix	Local operations and induced Kraus updates	Are the operations causally admissible and non-signalling?

This benchmark suggests a disciplined order for future comparisons. First match the single-probe probability after integrating over the detection time. Next match joint event statistics for two strictly spacelike supports. Only after an explicit probe instrument has been fixed should one compare the final detector density matrix and its entanglement or other resource witnesses. Finally, test causal composition and finite-apparatus reliability. This order prevents a field correlator, an event-probability hierarchy, a POVM effect, a quantum instrument, and a resource measure from being treated as if they were the same physical object.

The measurement frameworks above define idealized operations and outcome probabilities. Practical resource certification also depends on finite detector sensitivity, noise, postselection, and the control of switching and spatial support.

S1.1 Two instruments with the same effects

For Eq. (5), a Kraus operator of $\mathcal{I}_r$ is $K_r = I/\sqrt{2}$. For $\mathcal{J}_r$ use $L_{rj} = |0\rangle\langle j|/\sqrt{2}$, with $j = 0, 1$. Then

$$K_r^\dagger K_r = \sum_{j=0}^{1} L_{rj}^\dagger L_{rj} = \tfrac{1}{2} I, \qquad \sum_r E_r = I. \tag{S8}$$

Both maps are completely positive and trace non-increasing; summing over r yields trace-preserving channels. For a reference R and input $\Phi_{RS}^+ = |\Phi^+\rangle\langle\Phi^+|$, the normalized conditional outputs are

$$\rho_{RS|r}^{\mathcal{I}} = \Phi_{RS}^+, \qquad \rho_{RS|r}^{\mathcal{J}} = \tfrac{1}{2} I_R \otimes |0\rangle\langle 0|_S. \tag{S9}$$

The conventional negativity $(\|\rho^{T_S}\|_1 - 1)/2$ equals $1/2$ and 0, respectively. In both cases $p_r = 1/2$. This proves the distinction in the main text with local finite-dimensional channels, without asserting their equivalence as measurements of a continuum field.

More generally, if $\mathcal{E}_S = \sum_r \mathcal{I}_r$ is trace preserving, then $\mathcal{E}_S^*(I_S) = I_S$. For any observable B of an untouched finite system,

$$\mathrm{Tr}[(I_S \otimes B)(\mathcal{E}_S \otimes \mathrm{id})(\rho_{SB})] = \mathrm{Tr}[(\mathcal{E}_S^*(I_S) \otimes B)\rho_{SB}] = \mathrm{Tr}[(I_S \otimes B)\rho_{SB}]. \tag{S10}$$

All expectation values of the nonselective B marginal are unchanged, proving Eq. (6). QFT implementations additionally require a causally admissible localized coupling; the finite-system calculation alone does not establish that requirement.

S2 Finite-mode transfer and channel dependence

S2.1 A finite-mode resource-transfer benchmark

A finite-mode model provides a transparent test of the field-to-detector bounds before one attempts a continuum-field calculation. Consider two bosonic modes in a two-mode squeezed vacuum,

$$|\Psi_r\rangle_{\mathrm{F}} = \sqrt{1-\zeta^2}\sum_{n=0}^{\infty}\zeta^n|n\rangle_A|n\rangle_B, \qquad \zeta = \tanh r, \tag{S11}$$

whose field mutual information is $I_{\mathrm{F}} = 2g(\sinh^2 r)$, with $g(n) = (n+1)\ln(n+1) - n\ln n$. Correlations between atoms and two-mode squeezed radiation have earlier been studied in generalized two-mode Jaynes–Cummings models; the present calculation specializes to two independent local resonant channels and evaluates the resulting detector state explicitly [6]. Each mode interacts locally with a two-level detector initially in its ground state. In the resonant Jaynes–Cummings model,

$$U_\nu(\theta) = \exp[-i\theta(\sigma_\nu^+ a_\nu + \sigma_\nu^- a_\nu^\dagger)], \qquad |g,n\rangle \mapsto \cos(\theta\sqrt{n})|g,n\rangle - i\sin(\theta\sqrt{n})|e,n-1\rangle, \tag{S12}$$

Here θ is the dimensionless resonant interaction angle, a_ν the mode annihilation operator, and $\sigma_\nu^\pm$ the detector ladder operators. The second term is absent for $n = 0$. Tracing out the field modes gives a two-qubit detector state. The Fock expansion is truncated at $N_{\max} = 30$; increasing the cutoff to 40 changes the reported quantities by less than the displayed precision over $0.05 \le r \le 0.8$. No postselection is used.

Figure S1 shows that the detector negativity depends nontrivially on the interaction angle. At $\theta = \pi/2$, representative values of the field and detector mutual informations and the detector negativity are listed in Table S1. They satisfy $I(D_A : D_B) \le I_{\mathrm{F}}$ and the two-qubit negativity bound. The table and angle scan are complementary: the former gives a fixed-angle comparison, while the latter shows how the transfer changes during the local interaction.

Table S1: Finite-mode Jaynes–Cummings resource-transfer benchmark at $\theta = \pi/2$. Mutual informations are in nats.

r	I_{F}	$I(D_A : D_B)$	$\mathcal{N}$	Upper bound
0.05	0.03499	0.03480	0.04983	0.26453
0.10	0.11251	0.11009	0.09866	0.47436
0.20	0.34258	0.31445	0.18936	0.50000
0.40	0.96490	0.69518	0.32103	0.50000
0.60	1.68845	0.84182	0.36842	0.50000
0.80	2.45470	0.78027	0.34684	0.50000

A sharper state-dependent estimate is obtained by applying data processing after only the first local interaction. For the normalized truncated Fock coefficients c_n, the state of detector

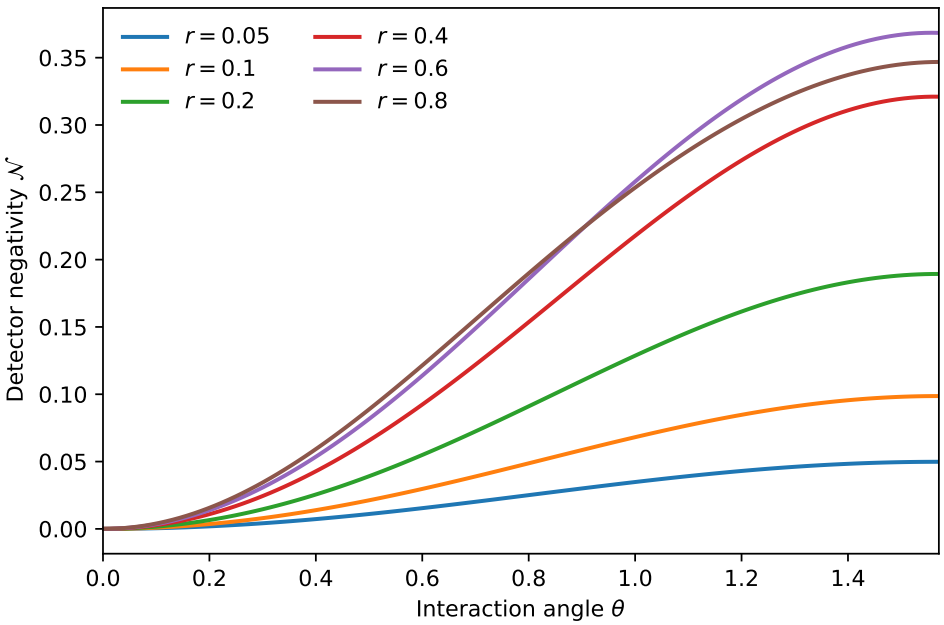


Figure S1: Detector negativity versus the local Jaynes–Cummings interaction angle for several initial squeezing parameters. Dashed lines show the two-qubit upper bound derived from the initial field mutual information. This finite-mode benchmark is not a spacelike-separated relativistic detector calculation.

D_A and the untouched mode F_B is

$$\rho_{D_A F_B} = \sum_{m=0}^{N_{\max}} |v_m\rangle\langle v_m|, \quad |v_m\rangle = c_m \cos(\theta\sqrt{m})|g,m\rangle - ic_{m+1}\sin(\theta\sqrt{m+1})|e,m+1\rangle, \quad \text{(S13)}$$

with $c_{N_{\max}+1} = 0$. Its mutual information is finite because D_A is a qubit. The one-sided bound in Eq. (S33) reduces the field-derived negativity bound by about 29% at $r = 0.05$ and 0.10 for $\theta = \pi/4$; at larger squeezing the two-qubit cap $1/2$ becomes active. This is a state-specific estimate and does not contradict the unit worst-case contraction coefficient of the unbounded JC channel.

For the two-mode squeezed input, the detector state has an exact X form. Define

$$a_n = \cos^2(\theta\sqrt{n}), \qquad b_n = \sin^2(\theta\sqrt{n}). \quad \text{(S14)}$$

In the basis $(|gg\rangle, |ge\rangle, |eg\rangle, |ee\rangle)$,

$$\rho_{D_A D_B} = \begin{pmatrix} P_{gg} & 0 & 0 & Z \\ 0 & P_{ge} & 0 & 0 \\ 0 & 0 & P_{ge} & 0 \\ Z^* & 0 & 0 & P_{ee} \end{pmatrix}, \quad \text{(S15)}$$

where

$$P_{gg} = (1-\zeta^2)\sum_{n=0}^{\infty}\zeta^{2n}a_n^2,$$
$$P_{ge} = (1-\zeta^2)\sum_{n=0}^{\infty}\zeta^{2n}a_n b_n,$$
$$P_{ee} = (1-\zeta^2)\sum_{n=0}^{\infty}\zeta^{2n}b_n^2, \tag{S16}$$
$$Z = -(1-\zeta^2)\zeta\sum_{n=0}^{\infty}\zeta^{2n}a_n b_{n+1}.$$

The partial transpose gives the exact negativity

$$\mathcal{N} = \max\left\{0,\,(1-\zeta^2)\left[\zeta\sum_{n=0}^{\infty}\zeta^{2n}a_n b_{n+1} - \sum_{n=0}^{\infty}\zeta^{2n}a_n b_n\right]\right\}. \tag{S17}$$

For weak squeezing, $\zeta = \tanh r \ll 1$,

$$\mathcal{N} = r\sin^2\theta + O(r^2), \qquad I_{\mathrm{F}} = 2r^2[1-\ln(r^2)] + O(r^4|\ln r|). \tag{S18}$$

The detector negativity is therefore linear in r at leading order, whereas the field mutual information is of order $r^2|\ln r|$. This explains why the general mutual-information bound is loose for weakly squeezed inputs.

There is also a bound based only on local detector populations. For any two-qubit state,

$$\mathcal{N}(\rho_{D_AD_B}) \le \sqrt{\det\rho_{D_A}}, \qquad \rho_{D_A} = \begin{pmatrix} 1-p_A & c_A \\ c_A^* & p_A \end{pmatrix}. \tag{S19}$$

For a pure two-qubit state with Schmidt eigenvalues $s, 1-s$, the negativity equals $\sqrt{s(1-s)} = \sqrt{\det\rho_{D_A}}$. Negativity is convex under mixtures. For a qubit marginal with Bloch vector $\boldsymbol{r}$, the function $\sqrt{\det\rho_{D_A}} = \frac{1}{2}\sqrt{1-|\boldsymbol{r}|^2}$ is concave on the Bloch ball (its interior Hessian is negative semidefinite, and the boundary follows by continuity). Applying these two properties to any pure-state decomposition of a mixed state proves Eq. (S19).

The JC interaction conserves $\hat{n}_A + |e_A\rangle\langle e_A|$, so $p_A \le \bar{n}_A$ and similarly for B. Hence

$$\mathcal{N} \le \min\{\nu(\bar{n}_A), \nu(\bar{n}_B)\}, \qquad \nu(x) = \sqrt{x_*(1-x_*)}, \quad x_* = \min\{x, \tfrac{1}{2}\}. \tag{S20}$$

This population bound is accessible from local measurements, but its excitation-conservation premise does not apply unchanged to the full Unruh–DeWitt interaction with counter-rotating terms.

The finite-mode benchmark confirms the internal consistency of the data-processing bounds and illustrates the dependence of resource transfer on the detector coupling. It is not a spacelike-separated relativistic harvesting calculation. A continuum-field extension must specify compact spacetime supports for the local modes and control the corresponding local-algebra mutual information.

S2.2 Contraction coefficients and mean-energy constraints

A strict data-processing contraction can sharpen the field-to-detector bound, but it must be established for the actual detector channel and input class. For a regulated channel $\mathcal{E}: A \to D$, define

$$\eta_{\mathrm{MI}}(\mathcal{E}) = \sup_{\rho_{RA}: I(R:A)>0} \frac{I(R:D)_{(\mathrm{id}_R\otimes\mathcal{E})(\rho)}}{I(R:A)_\rho}, \qquad 0 \le \eta_{\mathrm{MI}} \le 1. \tag{S21}$$

The product-reference coefficient relevant to ordinary mutual information is bounded by the corresponding complete conditional-mutual-information contraction coefficient [7]. For two local detector channels, sequential data processing gives

$$I(D_A : D_B) \leq \eta_{\mathrm{MI}}(\mathcal{E}_A)\eta_{\mathrm{MI}}(\mathcal{E}_B) I_{\mathrm{F}}(A : B), \tag{S22}$$

and hence, for two qubits,

$$\mathcal{N}(\rho_{D_A D_B}) \leq \min\left\{\frac{1}{2}, \sqrt{2\eta_A \eta_B I_{\mathrm{F}}}\right\}. \tag{S23}$$

For continuum local algebras, these coefficients must be defined with the same split or finite-resolution prescription used for I_{F}; no continuum value is assumed here.

The difficulty is that the ideal resonant Jaynes–Cummings (JC) channel need not contract mutual information uniformly. A detector initially in $|g\rangle$ maps a Fock input $|n\rangle$ to an excited-state probability

$$p_n = \sin^2(\theta\sqrt{n}). \tag{S24}$$

For the binary code $\{|0\rangle, |n\rangle\}$, the input mutual information is $\ln 2$ and the detector output mutual information is

$$I(U : D) = h_2(p_n/2) - \frac{1}{2} h_2(p_n). \tag{S25}$$

For every fixed $\theta > 0$, a subsequence has $p_n \to 1$. A direct construction avoids any equidistribution assumption: take $a_j = (\pi/2 + j\pi)/\theta$ and $n_j = \lfloor a_j^2 + 1/2 \rfloor$. Then

$$|\theta\sqrt{n_j} - (\pi/2 + j\pi)| \leq \frac{\theta}{2(\sqrt{n_j} + a_j)} \longrightarrow 0. \tag{S26}$$

Thus $\sin^2(\theta\sqrt{n_j}) \to 1$. The output retains asymptotically all of the equiprobable input bit, and data processing fixes the exact worst-case coefficient at

$$\eta_{\mathrm{MI}}(\mathcal{E}_\theta) = 1. \tag{S27}$$

This is a statement about the supremum over all Fock inputs, not about the transfer efficiency for a specified squeezed state.

An average photon-number constraint alone does not restore a strict coefficient. Encode a rare bit in $|n\rangle$ with probability $q_n = E/n$ and otherwise use $|0\rangle$. The mean photon number is E, while the decoding error probability is $q_n(1 - p_n)$. Along the subsequence with $p_n \to 1$, the conditional entropy of the bit given the detector is negligible compared with its input entropy. Therefore

$$\eta_{\mathrm{MI}}^{(E)}(\mathcal{E}_\theta) = 1, \qquad E > 0. \tag{S28}$$

For completeness, the exact information ratio of this binary classical code is

$$\frac{I(U : D)}{I(U : F)} = \frac{h_2(q_n p_n) - q_n h_2(p_n)}{h_2(q_n)} \longrightarrow 1, \tag{S29}$$

where $h_2(q) = -q \ln q - (1 - q)\ln(1 - q)$ uses natural logarithms. Along the constructed sequence, $q_n \to 0$ and $p_n \to 1$; the expansion $h_2(q) = q\ln(1/q) + O(q)$ gives the limit, while $q_n h_2(p_n)/h_2(q_n) \to 0$. This is closely related to the average-cost obstruction for classical channel contraction established by Polyanskiy and Wu [8]. An average photon-number bound alone is therefore insufficient. Hard spectral restrictions or additional channel noise may restore a strict coefficient, but each proposed restriction still needs its own channel analysis.

A more informative object is the nonlinear information-dissipation curve

$$\mathfrak{F}_E(s) = \sup_{\substack{I(R:F) \leq s \\ \mathrm{Tr}(\hat{n}\rho_F) \leq E}} I(R : D). \tag{S30}$$

It obeys $\mathfrak{F}_E(s) \le s$. At special angles the curve is exactly known on a finite interval: if $p_{n_*} = 1$, then

$$q_* = \min\left\{\frac{E}{n_*}, \frac{1}{2}\right\}, \qquad s_* = h_2(q_*), \qquad \mathfrak{F}_E(s) = s \quad (0 \le s \le s_*). \tag{S31}$$

For $\theta = \pi/4$ one may take $n_* = 4$, and for $\theta = \pi/2$ one may take $n_* = 1$. Away from such exact transitions, the rare-code argument establishes unit limiting slope near $s = 0$, but does not determine the full curve. To pass from the code sequence to this slope, note that $n_j \sim j^2$ in the construction above. Hence the input information values $s_j = h_2(E/n_j)$ satisfy $s_{j+1}/s_j \to 1$. Monotonicity of $\mathfrak{F}_E$ fills the intervals between them; combined with $\mathfrak{F}_E(s) \le s$ and Eq. (S29), it gives $\mathfrak{F}_E(s)/s \to 1$.

A strict bound can be recovered for a noisy readout. If the detector is followed by depolarizing noise of strength p, the composite channel contains a replacement component of weight p. The quantum Doeblin bound then gives

$$\eta_{\mathrm{MI}} \le 1 - p, \qquad I(D_A : D_B) \le (1 - p_A)(1 - p_B) I_{\mathrm{F}}. \tag{S32}$$

This result follows from the quantum Doeblin decomposition and the associated conditional-mutual-information contraction theorem [7, 9]. It applies to the stated regulated depolarizing-readout model, not automatically to amplitude damping, dephasing, dark counts, or an ideal continuum Unruh–DeWitt channel.

In this replacement-mixture case the bound also has a direct proof. Retain a classical flag F recording whether the output was replaced, with replacement probability p independent of the reference R. Before the flag is discarded, $I(R : DF) = (1-p) I(R : D)_{\mathrm{unreplaced}}$ because the replacement output is uncorrelated with R. Tracing out F cannot increase mutual information. Applying the same argument at the two readouts gives the product bound in Eq. (S32). This proof does not require assigning a contraction coefficient to an unregulated continuum channel.

Finally, a state-dependent estimate avoids the worst-case optimization. Applying data processing after only one local interaction yields

$$\begin{aligned} I(D_A : D_B) &\le \min\{I(D_A : B), I(A : D_B)\}, \\ \mathcal{N} &\le \min\left\{\frac{1}{2}, \sqrt{2 \min\{I(D_A : B), I(A : D_B)\}}\right\}. \end{aligned} \tag{S33}$$

Unlike a uniform channel coefficient, this bound retains information about the actual field state and detector coupling. The finite-mode application is given in Supplementary Sec. S2.

S3 Strictly spacelike pointlike detector limit

We now examine whether the entanglement extracted by two detectors survives when their spatial size tends to zero while their separation and switching duration remain fixed. Consider a massless real scalar field in four-dimensional Minkowski spacetime. The two identical UDW detectors are at rest, their centres are separated by L, and their spatial profiles are normalized, spherically symmetric functions $F_\sigma(\mathbf{x})$ supported in balls of radius σ. The switching is

$$\chi(t) = \begin{cases} (1 - t^2/T^2)^6, & |t| \le T, \\ 0, & |t| > T, \end{cases} \qquad \widetilde{\chi}(\omega) = \int_{-\infty}^{\infty} dt\, \chi(t) e^{i\omega t}. \tag{S34}$$

This switching is compactly supported and has five continuous derivatives at the endpoints. We use the smooth compact spatial profile

$$F_\sigma(\mathbf{x}) = C_\sigma \exp\left[-\frac{1}{1 - |\mathbf{x}|^2/\sigma^2}\right] \quad (|\mathbf{x}| < \sigma), \tag{S35}$$

and set it to zero outside the ball; C_σ is fixed by $\int d^3x\, F_\sigma(\mathbf{x}) = 1$. The full interaction regions are strictly spacelike separated provided

$$L - 2\sigma > 2T. \tag{S36}$$

At leading order in the detector–field coupling λ, the local excitation probability and the nonlocal coherence are

$$P_\sigma = \frac{\lambda^2}{4\pi^2}\int_0^\infty dk\, k\, |\widetilde{F}_\sigma(k)|^2 |\widetilde{\chi}(\Omega+k)|^2, \tag{S37}$$

$$\mathcal{M}_\sigma = -\frac{\lambda^2}{4\pi^2 L}\int_0^\infty dk\, \sin(kL) |\widetilde{F}_\sigma(k)|^2 \widetilde{\chi}(\Omega-k)\widetilde{\chi}(\Omega+k), \tag{S38}$$

where $\widetilde{F}_\sigma(k)$ is the radial Fourier transform of the spatial profile. Because the two supports are strictly spacelike separated, the Wightman function is smooth on the product of the two interaction regions, and the nonlocal integral has no light-cone singularity. The local response is also finite in the pointlike limit because the compact switching has sufficient differentiability for $\widetilde{\chi}(\omega)$ to decay rapidly. The leading-order negativity for identical detectors is

$$\mathcal{N}_\sigma^{(2)} = \max\{0, |\mathcal{M}_\sigma| - P_\sigma\}. \tag{S39}$$

For each fixed σ, a valid regular expansion gives $\mathcal{N}_\sigma = \mathcal{N}_\sigma^{(2)} + O(\lambda^4)$ in this zero-mean Gaussian model. No bound uniform in σ on that remainder is established here. All pointlike limits below therefore concern the second-order coefficient, rather than an exchange of the exact finite-coupling and pointlike limits.

The small-σ expansion follows from the second spatial moment. For a normalized isotropic profile,

$$\widetilde{F}_\sigma(k) = 1 - \frac{k^2\langle r^2\rangle_\sigma}{6} + O(k^4\sigma^4), \qquad \langle r^2\rangle_\sigma = \mu_2\sigma^2. \tag{S40}$$

Consequently, provided the switching makes the corresponding spectral moments convergent,

$$\frac{P_\sigma}{\lambda^2} = p_0 - \frac{\mu_2\sigma^2}{12\pi^2}\int_0^\infty dk\, k^3 |\widetilde{\chi}(\Omega+k)|^2 + O(\sigma^4), \tag{S41}$$

$$\frac{\mathcal{M}_\sigma}{\lambda^2} = m_0 + \frac{\mu_2\sigma^2}{12\pi^2 L}\int_0^\infty dk\, k^2 \sin(kL)\widetilde{\chi}(\Omega-k)\widetilde{\chi}(\Omega+k) + O(\sigma^4), \tag{S42}$$

where p_0 and m_0 denote the pointlike limits of P_σ/λ^2 and $\mathcal{M}_\sigma/\lambda^2$, respectively. Thus both detector response coefficients have finite pointlike limits for this switching class. If $|m_0| > p_0$, the leading-order negativity also has a strictly positive pointlike limit; if $|m_0| < p_0$, it vanishes in a neighbourhood of the pointlike limit. The equality case requires higher-order analysis.

For a concrete test, we set $T = 1$, $\Omega T = 9$, and $L/T = 2.033333\ldots$. The profile in Eq. (S35) has dimensionless second moment $\mu_2 = 0.33508696\ldots$. The spacelike margin is positive throughout the scan, including the largest radius $\sigma/T = 0.016$. The Fourier integrals were evaluated with a radial quadrature for the compact profile and a Gauss–Legendre quadrature for the switching transform. An independent reconstruction supplied in `reproduce_compact_detector.py` instead uses the analytic transform $\widetilde{\chi}(\omega) = 92160\, j_6(|\omega|)/|\omega|^6$ for $T = 1$, with its regular limit at zero. Comparing spectral cutoffs $k_{\max} = 300, 500$ and radial quadrature orders $96, 160$ changes the response coefficients by less than 2.5×10^{-17} and agrees with the archived scan to 1.6×10^{-17}. These differences are numerical convergence diagnostics, not certified error bounds. The pointlike values are

$$p_0 = 2.13325790\times10^{-6}, \qquad |m_0| = 3.57653182\times10^{-6}, \tag{S43}$$

so that

$$\lim_{\sigma\to0}\frac{\mathcal{N}_\sigma^{(2)}}{\lambda^2} = 1.44327392\times10^{-6}. \tag{S44}$$

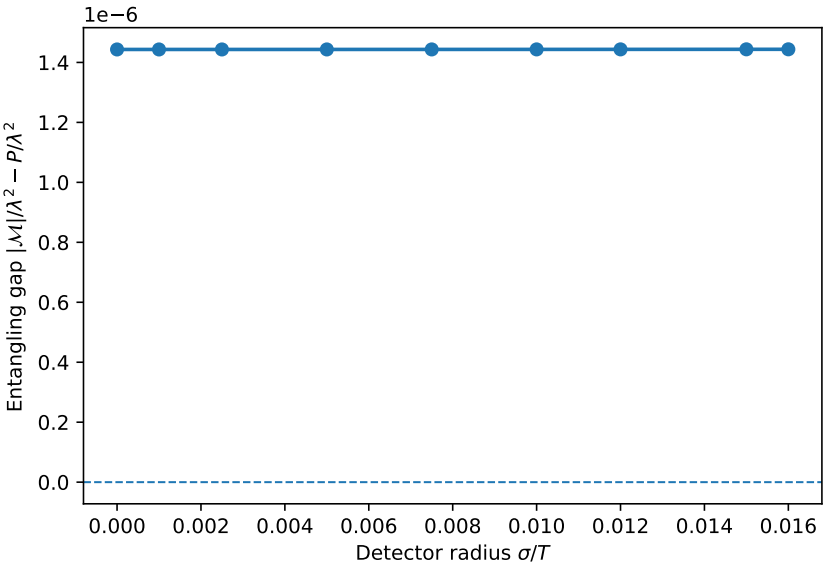


Figure S2: Leading entangling gap $|\mathcal{M}_\sigma|/\lambda^2 - P_\sigma/\lambda^2$ versus σ/T, for $\Omega T = 9$, $L/T = 61/30$, and the compact profiles of Eqs. (S34)–(S35). The full supports obey $L - 2\sigma > 2T$ throughout. The positive coefficient is a perturbative result; numerical convergence does not bound the omitted dynamical terms.

The fitted small-size expansion is

$$\frac{|\mathcal{M}_\sigma| - P_\sigma}{\lambda^2} = 1.44327392 \times 10^{-6} + 1.96659 \times 10^{-6} \left(\frac{\sigma}{T}\right)^2 + O\big((\sigma/T)^4\big)\,. \tag{S45}$$

The positive limit is therefore not caused by a finite spatial radius: it remains after the compact spatial profile is taken to a point. The local response, nonlocal coherence, and leading negativity approach finite pointlike limits. The positive entangling gap is shown directly in Fig. S2; the separate response and negativity plots are omitted because their limiting values are already stated explicitly above.

This result answers only the detector-side pointlike-limit question for the specified massless scalar model and switching. It does not establish the behaviour of the mutual information between shrinking field regions. In particular, the detector interaction samples a spacetime world tube of fixed temporal duration, whereas the mutual information of two equal-time spatial balls refers to different local regions. These quantities must not be identified. A separate algebraic or finite-resolution calculation is needed to determine the field-region mutual information and to compare its limit with the finite detector negativity. The present result also does not remove the need to keep the coupling perturbative: λ must be small enough that higher-order terms do not overturn the positive leading-order gap.

S4 Preparation-family contextuality and channel invertibility

S4.1 Transport of a preparation-family noncontextual model

Let $\mathcal{S} = \{\rho_s\}$ be a specified family of finite-dimensional preparations. In the model used for the channel argument of Ref. [10], there are Hermitian operators G_τ and density operators σ_τ such that

$$\mathrm{Tr}(\rho_s M) = \sum_\tau \mu_s(\tau)\,\mathrm{Tr}(\sigma_\tau M), \quad \mu_s(\tau) = \mathrm{Tr}(\rho_s G_\tau) \geq 0, \quad \sum_\tau \mu_s(\tau) = 1 \tag{S46}$$

for every s and every POVM effect M. Positivity of G_τ on every quantum state is not assumed; its weights must be nonnegative on the specified preparation family. An integral may replace the sum for a continuous ontic space. The linear assignment enforces the specified preparation equivalences and mixtures; the Born response enforces measurement equivalences.

Proposition (linear-invertible channel transport). If Λ is a common CPTP map, invertible on the full real vector space of Hermitian operators, then the output family $\{\Lambda(\rho_s)\}$ admits a representation of the same form.

Proof. Let Λ^* be the trace-dual map and set

$$G'_\tau = (\Lambda^{-1})^*(G_\tau), \qquad \sigma'_\tau = \Lambda(\sigma_\tau). \tag{S47}$$

Then $\mathrm{Tr}[\Lambda(\rho_s)G'_\tau] = \mu_s(\tau)$, so the weights stay nonnegative and normalized on the output family. For any effect M,

$$\begin{aligned}\sum_\tau \mu_s(\tau)\,\mathrm{Tr}(\sigma'_\tau M) &= \sum_\tau \mu_s(\tau)\,\mathrm{Tr}[\sigma_\tau \Lambda^*(M)] \\ &= \mathrm{Tr}[\rho_s \Lambda^*(M)] = \mathrm{Tr}[\Lambda(\rho_s)M].\end{aligned} \tag{S48}$$

The adjoint of a CPTP map sends POVMs to POVMs, so Eq. (S46) applies in the second step. The σ'_τ remain density operators. Finally, invertibility implies that a linear relation among output preparations holds exactly when the corresponding relation holds among the inputs. This proves transport of the specified representation and its operational equivalences. No physical inverse channel was used.

The conclusion is conditional on this family, its equivalences, and the common channel. It does not identify the representation with the contextual fraction of an arbitrary fixed compatible-measurement empirical model. It is not a statement that one density operator is intrinsically noncontextual independent of the preparations and measurements allowed.

S4.2 Exact weak-coupling invertibility and a limiting pitfall

For a finite-dimensional exact channel Λ_λ with $\Lambda_0 = \mathrm{id}$, continuity in any submultiplicative superoperator norm implies invertibility whenever $\|\Lambda_\lambda - \mathrm{id}\| < 1$. Indeed, the convergent Neumann series is

$$\Lambda_\lambda^{-1} = \sum_{r=0}^{\infty} (-1)^r (\Lambda_\lambda - \mathrm{id})^r. \tag{S49}$$

This establishes a sufficiently small coupling interval, without specifying its size for an unbounded field interaction. A truncated Dyson expression is not automatically CPTP; its formal inverse alone does not establish the proposition's physical assumptions.

Invertible approximants may have a noninvertible limit. For example, let $\mathcal{D}$ remove the off-diagonal entries of a qubit state in a fixed basis, and define

$$\Lambda_n = \frac{1}{n}\,\mathrm{id} + \left(1 - \frac{1}{n}\right)\mathcal{D} \qquad (n = 1, 2, \ldots). \tag{S50}$$

Every Λ_n is CPTP and invertible as a linear map: its off-diagonal eigenvalue is $1/n$. Its norm limit $\mathcal{D}$ has a nontrivial kernel. Thus invertibility at every finite approximation cannot justify a claim at arbitrary finite coupling without a uniform lower bound on the relevant eigenvalues or singular values. This qualifies the extrapolation discussed in Ref. [10]; it does not invalidate its conditional channel proposition or exact gapless construction.

S5 Resource definitions, local channels, and perturbative diagnostics

The finite-state examples below distinguish a resource definition, a structural channel identity, and a source-specific perturbative diagnostic. We use $N = 2\mathcal{N}$ throughout this section. The companion checker constructs the counterexamples and the cluster channels directly, and reproduces the displayed W/Dicke expressions. No exact finite-coupling revival or all-orders remainder is established.

S5.1 Mixed-state GME and residual negativity

The approximation issue is logically separate from the definition of genuine multipartite entanglement (GME). A mixed state is biseparable if it can be written as a convex mixture of states that are separable across possibly different bipartitions; GME means that no such decomposition exists. This convex definition is why mixed-state certification cannot in general be inferred from a residual of pairwise negativities alone. General separability criteria and witnesses have been developed specifically for GME [11], and reviews of multipartite measures emphasize monotonicity, mixed-state extensions, and operational meaning as independent requirements [12].

Residual-negativity constructions are historically motivated by negativity monogamy [13], but the raw minimum residual is not a faithful GME certificate on arbitrary mixed states. The explicitly biseparable state

$$\rho_{\mathrm{bisep}} = \frac{1}{2}|\Phi^+\rangle_{AB}\langle\Phi^+| \otimes |0\rangle_C\langle 0| + \frac{1}{2}|\Phi^+\rangle_{AC}\langle\Phi^+| \otimes |0\rangle_B\langle 0|, \tag{S51}$$

has residual squared negativities approximately 0.3090, 0.1545, and 0.1545, despite being biseparable by construction. We therefore call this quantity a *minimum residual-negativity indicator* unless an independent mixed-state GME criterion is supplied.

The four-qubit Dicke residual π_4 has an analogous and even sharper limitation. Consider

$$\begin{aligned} \rho^{(4)}_{\mathrm{bisep}} &= \frac{1}{3}\left(\rho_{AB|CD} + \rho_{AC|BD} + \rho_{AD|BC}\right), \\ \rho_{XY|ZW} &= \left(|\Phi^+\rangle\langle\Phi^+|\right)_{XY} \otimes \left(|\Phi^+\rangle\langle\Phi^+|\right)_{ZW}. \end{aligned} \tag{S52}$$

Each component is separable across a $2|2$ bipartition, so the mixture is biseparable. Nevertheless all four one-versus-three negativities equal one, all six two-qubit negativities vanish, and the residual definition used in the Dicke analysis gives

$$\pi_A = \pi_B = \pi_C = \pi_D = \pi_4 = 1. \tag{S53}$$

The values can be obtained analytically. For the $A|BCD$ cut, write the three pure components as

$$\begin{aligned} |\psi_i\rangle &= \frac{|0\rangle|u_i\rangle + |1\rangle V|u_i\rangle}{\sqrt{2}}, \\ |u_i\rangle &\in \left\{\frac{|000\rangle + |011\rangle}{\sqrt{2}}, \frac{|000\rangle + |101\rangle}{\sqrt{2}}, \frac{|000\rangle + |110\rangle}{\sqrt{2}}\right\}, \end{aligned} \tag{S54}$$

Here $X = |0\rangle\langle 1| + |1\rangle\langle 0|$ and $V = X \otimes X \otimes X$ flips the three bits. The u_i have even parity and Vu_i odd parity. Their equal Gram matrices permit a local isometry on BCD taking u_i to $|0\rangle \otimes \xi_i$ and Vu_i to $|1\rangle \otimes \xi_i$. The mixture is thus locally equivalent to a Bell state tensored with $\tau = \frac{1}{3}\sum_i |\xi_i\rangle\langle\xi_i|$. The Gram matrix has diagonal entries one and off-diagonal entries $1/2$, so τ has nonzero eigenvalues $2/3, 1/6, 1/6$. The negative eigenvalues of the full partial transpose are consequently $-1/3, -1/12, -1/12$, giving $N_{A|BCD} = 1$. Permutation symmetry

gives the other three cuts. Every pair marginal equals $\frac{1}{3}\Phi^+ + \frac{2}{3}I_4/4$, whose partial transpose has smallest eigenvalue zero, hence zero negativity. This proves the stated residual values without a numerical entanglement inference.

Thus a maximal raw π_4 does not, by itself, certify four-partite GME for an arbitrary mixed state. In this review we therefore use the term *residual* π_4 *indicator* when discussing the field-traced detector state, unless a separate GME witness, convex-roof construction, or proven mixed-state monotone is provided.

S5.2 Exact local-channel invariance

Recent work has examined whether the structure of an initial multipartite state can protect selected entanglement measures against acceleration. A useful example is the four-qubit cluster state

$$|\mathrm{CL}_4\rangle = \frac{1}{2}\big(|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle\big), \tag{S55}$$

for which Li *et al.* report that the negativities across the $A|BCD$ and $B|ACD$ partitions remain maximal when detector D accelerates and interacts locally with a scalar field [14]. The result is partition specific: the partitions involving C or the accelerated detector D do not obey the same invariance.

For the $A|BCD$ and $B|ACD$ partitions, the freezing admits a stronger structural explanation than the first-order calculation alone suggests. Tracing detector D from Eq. (S55) already gives

$$N_{A|BC}(\rho_{ABC}) = N_{B|AC}(\rho_{ABC}) = 1, \qquad \rho_{ABC} = \mathrm{Tr}_D\,|\mathrm{CL}_4\rangle\langle\mathrm{CL}_4|, \tag{S56}$$

Indeed, $\rho_{ABC} = \frac{1}{2}\Phi^+_{AB}\otimes|0\rangle_C\langle 0| + \frac{1}{2}\Phi^-_{AB}\otimes|1\rangle_C\langle 1|$, where $|\Phi^-\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$ and $\Phi^- = |\Phi^-\rangle\langle\Phi^-|$. The orthogonal flag on C leaves a Bell state in each block; each partially transposed block has trace norm two before its weight is applied. This gives the two normalized negativities in Eq. (S56), using $N = \|\rho^{T_A}\|_1 - 1$. Suppose the field is initially uncorrelated with the detectors and only detector D couples to it. The exact reduced detector dynamics then defines a local completely positive trace-preserving map Λ_D on D. Trace preservation implies

$$\mathrm{Tr}_D[(I_{ABC}\otimes\Lambda_D)(\rho_{ABCD})] = \mathrm{Tr}_D\,\rho_{ABCD}. \tag{S57}$$

Partial trace is a local operation on the BCD side, so negativity monotonicity gives

$$N_{A|BCD}[(I_{ABC}\otimes\Lambda_D)(\rho)] \geq N_{A|BC}(\mathrm{Tr}_D\,\rho) = 1. \tag{S58}$$

Because the first subsystem is a qubit, the same negativity is bounded above by one. Hence

$$N_{A|BCD} = 1, \qquad N_{B|ACD} = 1, \tag{S59}$$

for *any* local CPTP channel on D, not only the perturbative Unruh–DeWitt channel. We verified this numerically for amplitude-damping and dephasing channels over their full parameter ranges. This observation strengthens the freezing statement but changes its interpretation: for these two partitions, the invariance is a state-and-channel structural property rather than evidence for a uniquely acceleration-induced protection mechanism.

The perturbative range remains relevant for other cluster-state quantities. In the published detector calculation, with $q = \exp(-2\pi\Omega/a)$ and effective coupling ν, the norm of the first-order state has the correction

$$\|\Psi\|^2 = 1 + \delta_{\mathrm{CL}}, \qquad \delta_{\mathrm{CL}} = \frac{\nu^2(1+q)}{2(1-q)}. \tag{S60}$$

At fixed nonzero ν^2, δ_{CL} is nonuniform as $q \to 1$. This does not undermine the exact argument in Eq. (S59), but it does limit conclusions about the detailed detector state, the $C|ABD$ and $D|ABC$ partitions, or task metrics derived from the truncated state in the infinite-acceleration limit.

S5.3 Nonuniform perturbative expressions

For reproducibility, the first-order W/Dicke diagnostic uses the following explicit state model, with D the interacting qubit:

$$\widetilde{\rho}_q = \rho + \frac{\nu^2}{1-q}\left(\sigma_D^- \rho \sigma_D^+ + q\sigma_D^+ \rho \sigma_D^-\right), \qquad \rho_q^{\text{test}} = \frac{\widetilde{\rho}_q}{\operatorname{Tr}\widetilde{\rho}_q}. \tag{S61}$$

The operators include the identity on the other qubits. The input is the equal superposition of all four-qubit computational states of Hamming weight one (W) or two (Dicke). This model reproduces the stated first-order normalization corrections and turning points. Its state-dependent normalization is not an exact CPTP evolution and cannot justify continuation to $q \to 1$. Small normalization correction is a necessary consistency diagnostic for retaining the unperturbed state as the leading term, not a sufficient bound on the dynamical error. In particular, squaring a first-order state retains the jump term while omitting second-order no-jump contributions of the same order in the reduced state. Normalization does not reconstruct those missing terms.

The same control test materially changes the interpretation of the reported W-state revival. For the four-qubit W state of Li *et al.* [15], the first-order correction is

$$\delta_W = \frac{\nu^2(1+3q)}{4(1-q)}. \tag{S62}$$

At $\nu^2 = 0.04$, the first-order expression has its minimum at $q \simeq 0.982236$, where $\delta_W \simeq 2.22$. By comparison, $\delta_W < 0.1$ requires $q < 0.6923$, and even $\delta_W < 1$ requires $q < 0.9612$. Appendix B of the source also gives a second-order density-matrix truncation. Reconstructing Eqs. (B7)–(B8), we find a minimum of its formal partial-transpose trace norm near $q \simeq 0.94427$, where $\nu^2/(1-q) \simeq 0.718$ is not small. The truncated matrix itself has a negative eigenvalue of approximately -2.4×10^{-4} there. The problem can be seen algebraically: its compression to $|0001\rangle$ and $(|0010\rangle + |0100\rangle + |1000\rangle)/\sqrt{3}$ is

$$\begin{pmatrix} k_1 & \sqrt{3}\,k_2 \\ \sqrt{3}\,k_2 & 3k_3 \end{pmatrix}, \qquad \begin{aligned} k_1 &= \frac{1-q-\nu^2}{4(1-q)}, \\ k_2 &= \frac{2(1-q)-(1+q)\nu^2}{8(1-q)}, \\ k_3 &= \frac{1-q-q\nu^2}{4(1-q)}. \end{aligned} \tag{S63}$$

Its determinant is $3(k_1k_3 - k_2^2) = -3\nu^4/64 < 0$ for nonzero ν. Thus even a small negative eigenvalue of this truncation is expected at the omitted order; it is not by itself a disproof of perturbation theory. However, the matrix is not an exact density operator, and its trace-norm expression must not be interpreted as a physical entanglement monotone outside a controlled approximation. In particular, the high-q branch is a property of the displayed truncated expressions; those expressions alone do not establish an exact finite-coupling revival.

The Dicke-state revival shows the same issue even though the coupling used in that work is smaller. Wu *et al.* consider a four-qubit Dicke state with $\nu^2 = 0.01$ and report non-monotonic behavior of both $N_{A(BCD)}$ and the residual quantity π_4 [16]. Their first-order normalization correction is

$$\delta_D = \frac{\nu^2(1+q)}{2(1-q)}. \tag{S64}$$

Reconstructing the published reduced density matrix, we obtain the minimum of $N_{A(BCD)}$ at $q \simeq 0.992912$ and the minimum of the reported π_4 at essentially the same point. There $\delta_D \simeq 1.406$. The conservative requirement $\delta_D < 0.1$ gives $q < 0.9048$, while $\delta_D < 1$ gives $q < 0.99005$; at the value $q = 0.99$ used in the paper's coupling scan, $\delta_D \simeq 0.995$. Thus the upward branch begins only once the first-order correction is already order unity.

S6 A separated composite-observable kernel

S6.1 A separated energy-density kernel must retain the distance

Consider a massless real scalar field in the $1+1$-dimensional Minkowski vacuum and the normal-ordered energy density $\mathcal{O}(t,x) =: T_{00}(t,x) := \frac{1}{2} :[(\partial_t\phi)^2 + (\partial_x\phi)^2]:$. For two inertial worldlines at $x_A = 0$ and $x_B = L > 0$, put $u = t - t'$. Wick contraction, or differentiation of the left- and right-moving logarithmic two-point functions, gives

$$\langle \mathcal{O}(t,0)\mathcal{O}(t',L)\rangle = \frac{1}{8\pi^2}\left[\frac{1}{(u-L-i0)^4} + \frac{1}{(u+L-i0)^4}\right]. \tag{S65}$$

To see the coefficient, use null coordinates $v_\pm = t \pm x$, with $\langle \phi_\pm(v)\phi_\pm(v')\rangle = -(4\pi)^{-1}\ln(v-v'-i0)$ up to an irrelevant constant. Each derivative correlator is $-(4\pi)^{-1}(v-v'-i0)^{-2}$. Since $T_{00} = (\partial_{v_+}\phi)^2 + (\partial_{v_-}\phi)^2$, the two Wick contractions in each independent chiral sector yield the two terms in Eq. (S65). The infrared constant drops out under differentiation.

Only at $L = 0$ does this become $[4\pi^2(u-i0)^4]^{-1}$, singular at $u = 0$. For $L > 0$, the cross-kernel's singular support lies at $u = \pm L$. In particular, for smooth compact switchings on $[-T, T]$ with $L > 2T$, it is smooth on the whole interaction domain. On this strictly spacelike domain, the time-ordering branches agree by microcausality and the cross integral is finite. The local response still requires its own distributional treatment.

This check matters when using Ref. [17]. Its local pointlike kernel, Eq. (35), is used in the displayed nonlocal reduction, Eq. (44), without an explicit separation dependence. That reduction has the coincident-position form and cannot be applied unchanged to arbitrary separated worldlines. The spatially smeared analysis there retains a separation-dependent convolution and is a different calculation. We therefore report its Gaussian-smearing and switching results with their hypotheses, and do not infer a universal separated-pointlike divergence criterion from the time-only expression. The example above is a regularity check, not a demonstration of positive harvested negativity.

For a general composite observable with nonzero mean $m(x) = \langle \mathcal{O}(x)\rangle$, the decomposition $\mathcal{O} = mI + \delta\mathcal{O}$ also separates a local deterministic drive from fluctuations. In a local interaction picture for that drive, the leading fluctuation calculation uses $\langle \delta\mathcal{O}(x)\delta\mathcal{O}(x')\rangle$. Mean-induced coherences must be accounted for before applying a zero-mean partial-transpose block. A zero mean of the underlying scalar field does not imply $m = 0$ for its energy density. These bookkeeping and ultraviolet checks are independent of the test for signalling between complete supports.

S7 Finite-sample certification with a calibrated witness

This section makes the distinction between an ideal-state witness and an experimental certificate quantitative. It uses a standard concentration inequality, not a new resource measure or an analysis of experimental raw data.

Let $W_{\mathrm{ent}} = w_0 I + \sum_{k=1}^{K} w_k O_k$ be a fixed Hermitian witness, with real coefficients w_k, such that $\mathrm{Tr}(W_{\mathrm{ent}}\sigma) \geq 0$ for every separable state σ. In setting k, an ideal measurement of O_k has outcomes in $[a_k, b_k]$. The n_k observed outcomes X_{ki} are independent, bounded in that interval, and have the same mean within each setting. All settings address the same preparation ρ. Define

$$\overline{X}_k = \frac{1}{n_k}\sum_{i=1}^{n_k} X_{ki}, \qquad \widehat{w} = w_0 + \sum_{k=1}^{K} w_k \overline{X}_k. \tag{S66}$$

Calibration is represented separately by a bound

$$\left|\mathrm{Tr}(W_{\mathrm{ent}}\rho) - \left(w_0 + \sum_{k=1}^{K} w_k \mathbb{E}X_{ki}\right)\right| \leq \Delta_{\mathrm{cal}}. \tag{S67}$$

For example, if the actual outcome observable is $\widetilde{O}_k$ and independently established operator-norm bounds satisfy $\|\widetilde{O}_k - O_k\|_\infty \le \epsilon_k$, one may take $\Delta_{\mathrm{cal}} = \sum_k |w_k|\epsilon_k$. State-preparation drift or correlated noise is not automatically included in this model.

Proposition (a sufficient finite-sample certificate). For $\delta \in (0,1)$, let

$$\Delta_{\mathrm{stat}} = \sum_{k=1}^{K} |w_k|(b_k - a_k)\sqrt{\frac{\ln(2K/\delta)}{2n_k}}. \tag{S68}$$

With probability at least $1-\delta$, the true witness expectation is no larger than $\widehat{w} + \Delta_{\mathrm{stat}} + \Delta_{\mathrm{cal}}$. A negative value of this upper bound therefore excludes separability under the stated sampling and calibration assumptions.

Proof. Hoeffding's inequality for bounded independent observations gives [18]

$$\Pr\big(|\overline{X}_k - \mathbb{E}X_{ki}| \ge t_k\big) \le 2\exp\left[-\frac{2n_k t_k^2}{(b_k - a_k)^2}\right]. \tag{S69}$$

Taking $t_k = (b_k - a_k)\sqrt{\ln(2K/\delta)/(2n_k)}$ makes each failure probability at most δ/K. The union bound then makes all K deviations hold simultaneously with probability at least $1-\delta$. The triangle inequality bounds the deviation of their weighted sum by $\sum_k |w_k| t_k = \Delta_{\mathrm{stat}}$. Adding Eq. (S67) proves the claim. If the calibration bound is itself statistical with failure probability δ_{cal}, the total confidence is at least $1 - \delta - \delta_{\mathrm{cal}}$, without requiring independence of the two bounds. A deterministic calibration bound must not be silently substituted for an uncertain calibration fit.

A two-qubit example. For $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, a witness is

$$W_\Phi = \frac{1}{2}I - |\Phi^+\rangle\langle\Phi^+| = \frac{1}{4}(I - X\otimes X + Y\otimes Y - Z\otimes Z), \tag{S70}$$

where X, Y, Z are Pauli matrices. The squared overlap of a pure product state with $|\Phi^+\rangle$ is at most $1/2$, by the Cauchy–Schwarz inequality; convexity gives the same bound for all separable states, proving the witness property. Each measured product has outcomes ± 1. With n repetitions for each of the three correlators,

$$\Delta_{\mathrm{stat}} = \frac{3}{2}\sqrt{\frac{\ln(6/\delta)}{2n}}. \tag{S71}$$

For an observed negative margin $m = -\widehat{w} > \Delta_{\mathrm{cal}}$, the sufficient condition becomes

$$n > \frac{9\ln(6/\delta)}{8(m - \Delta_{\mathrm{cal}})^2}. \tag{S72}$$

For illustration, $m = 0.125$, $\Delta_{\mathrm{cal}} = 0.020$, and $\delta = 0.05$ require $n \ge 489$ repetitions per correlator with this conservative bound. These are hypothetical inputs, not an experimental claim or a predicted harvesting yield. The guarantee is frequentist coverage of the fixed confidence rule; it is not a posterior probability that a particular state is entangled. If $m \le \Delta_{\mathrm{cal}}$, increasing n alone cannot certify that margin.

The witness and settings must be fixed independently of the data used in this bound, or selection must be accounted for. A predicted detector state with unknown higher-order corrections is a different uncertainty problem: more shots do not bound those corrections. Likewise, electro-optic quadratures are unbounded. Clipping them changes the measured observable, so any application needs an independently justified tail-error bound or a concentration inequality valid for their distribution. Coincidence covariance alone does not supply W_Φ or its calibration.

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